Integration by Parts

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The integration by parts formula**

Evaluate $\int xe^x\,dx$.

Let $u = x$, $dv = e^x\,dx$: $du = dx$, $v = e^x$. $\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C$.

2 **[Foundational]** • **Target: Definite integrals: area under a transcendental product**

Find the exact area under $y = xe^{-x}$ on $[0, 3]$, using $\int xe^{-x}\,dx = -xe^{-x} - e^{-x} + C$ (verify this antiderivative first by differentiating).

Check: $\frac{d}{dx}[-xe^{-x} - e^{-x}] = -e^{-x} + xe^{-x} + e^{-x} = xe^{-x}$ ✓. Area $= [-xe^{-x} - e^{-x}]_0^3 = (-3e^{-3} - e^{-3}) - (0 - 1) = 1 - 4e^{-3} \approx 1 - 0.199 \approx 0.801$.

3 **[Foundational]** • **Target: The integration by parts formula**

State the integration by parts formula, and explain the LIATE guideline for choosing u.

$\int u\,dv = uv - \int v\,du$. LIATE suggests choosing u in the priority order: Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential — the function type listed first should typically be u.

4 **[Foundational]** • **Target: The integration by parts formula**

Derive the integration by parts formula starting from the product rule $\frac{d}{dx}[uv] = u'v + uv'$.

Integrating both sides: $uv = \int u'v\,dx + \int uv'\,dx = \int v\,du + \int u\,dv$. Rearranging: $\int u\,dv = uv - \int v\,du$.

5 **[Foundational]** • **Target: The integration by parts formula**

Evaluate $\int x\cos x\,dx$.

Let $u = x$, $dv = \cos x\,dx$: $du = dx$, $v = \sin x$. $\int x\cos x\,dx = x\sin x - \int \sin x\,dx = x\sin x + \cos x + C$.

6 **[Foundational]** • **Target: Integration by parts with logarithms and inverse trig**

Evaluate $\int \ln x\,dx$.

Let $u = \ln x$, $dv = dx$: $du = \frac{1}{x}dx$, $v = x$. $\int \ln x\,dx = x\ln x - \int 1\,dx = x\ln x - x + C$.

7 **[Foundational]** • **Target: Integration by parts with logarithms and inverse trig**

Evaluate $\int x\ln x\,dx$.

Let $u = \ln x$, $dv = x\,dx$: $du = \frac{1}{x}dx$, $v = \frac{x^2}{2}$. $\int x\ln x\,dx = \frac{x^2}{2}\ln x - \int \frac{x}{2}\,dx = \frac{x^2}{2}\ln x - \frac{x^2}{4} + C$.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 **[Intermediate]** • **Target: Integration by parts with logarithms and inverse trig**

Evaluate $\int \tan^{-1} x\,dx$.

Let $u = \tan^{-1} x$, $dv = dx$: $du = \frac{1}{1+x^2}dx$, $v = x$.
$\int \tan^{-1} x\,dx = x\tan^{-1} x - \int \frac{x}{1+x^2}\,dx = x\tan^{-1} x - \frac{1}{2}\ln(1+x^2) + C$.

9 **[Intermediate]** • **Target: Integration by parts with logarithms and inverse trig**

Evaluate $\int x^2 \ln x\,dx$.

Let $u = \ln x$, $dv = x^2\,dx$: $du = \frac{1}{x}dx$, $v = \frac{x^3}{3}$. $\int x^2 \ln x\,dx = \frac{x^3}{3}\ln x - \int \frac{x^2}{3}\,dx = \frac{x^3}{3}\ln x - \frac{x^3}{9} + C$.

10 **[Intermediate]** • **Target: Integration by parts with logarithms and inverse trig**

Evaluate $\int \sin^{-1} x\,dx$.

Let $u = \sin^{-1} x$, $dv = dx$: $du = \frac{1}{\sqrt{1-x^2}}dx$, $v = x$.
$\int \sin^{-1} x\,dx = x\sin^{-1} x - \int \frac{x}{\sqrt{1-x^2}}\,dx = x\sin^{-1} x + \sqrt{1-x^2} + C$.

11 **[Intermediate]** • **Target: Repeated integration by parts**

Evaluate $\int x^2 e^x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = e^x\,dx$: $\int x^2 e^x\,dx = x^2e^x - 2\int xe^x\,dx$. Using the earlier result $\int xe^x\,dx = xe^x - e^x$:
$= x^2e^x - 2(xe^x - e^x) + C = x^2e^x - 2xe^x + 2e^x + C$.

12 **[Intermediate]** • **Target: Repeated integration by parts**

Evaluate $\int x^2 \sin x\,dx$, applying integration by parts twice.

First: $u = x^2$, $dv = \sin x\,dx$: $= -x^2\cos x + 2\int x\cos x\,dx$. Using the earlier result
$\int x\cos x\,dx = x\sin x + \cos x$: $= -x^2\cos x + 2x\sin x + 2\cos x + C$.

13 **[Intermediate]** • **Target: Repeated integration by parts**

Set up (do not evaluate fully) the reduction needed to find $\int x^3 e^x\,dx$, identifying how many times integration by parts must be applied.

Each application of IBP with $u = x^n$ reduces the power of x by 1 while v stays an antiderivative of e^x. Since the power starts at 3, integration by parts must be applied 3 times to eliminate the polynomial factor entirely.

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14 **[Intermediate]** • **Target: The cyclic case**

Evaluate $\int e^x \sin x \, dx$ using the cyclic (solve-for-the-integral) technique.

Let $I = \int e^x \sin x \, dx$. First IBP ($u = \sin x$, $dv = e^x dx$): $I = e^x \sin x - \int e^x \cos x \, dx$. Second IBP on the remaining integral ($u = \cos x$, $dv = e^x dx$): $\int e^x \cos x \, dx = e^x \cos x + \int e^x \sin x \, dx = e^x \cos x + I$. So $I = e^x \sin x - (e^x \cos x + I) \Rightarrow 2I = e^x \sin x - e^x \cos x \Rightarrow I = \frac{e^x(\sin x - \cos x)}{2} + C$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: The cyclic case** • **Cross-Lesson: 4.01 Polar Curves and Derivatives**

Evaluate $\int e^{2x} \cos x \, dx$ using the cyclic technique.

Let $I = \int e^{2x} \cos x \, dx$. IBP ($u = \cos x$, $dv = e^{2x} dx$): $I = \frac{1}{2} e^{2x} \cos x + \frac{1}{2} \int e^{2x} \sin x \, dx$. IBP again on that integral ($u = \sin x$, $dv = e^{2x} dx$): $\int e^{2x} \sin x \, dx = \frac{1}{2} e^{2x} \sin x - \frac{1}{2} I$. Substituting: $I = \frac{1}{2} e^{2x} \cos x + \frac{1}{4} e^{2x} \sin x - \frac{1}{4} I \Rightarrow \frac{5}{4} I = \frac{1}{2} e^{2x} \cos x + \frac{1}{4} e^{2x} \sin x \Rightarrow I = \frac{e^{2x}(2\cos x + \sin x)}{5} + C$.

16 **[Challenging]** • **Target: Definite integrals using integration by parts** • **Cross-Lesson: 4.02 Area in Polar Coordinates**

Evaluate $\int_0^1 x e^x \, dx$ using the antiderivative found earlier.

$[x e^x - e^x]_0^1 = (e - e) - (0 - 1) = 0 - (-1) = 1$.

17 **[Challenging]** • **Target: Definite integrals using integration by parts** • **Cross-Lesson: 5.01 Sequences and Basic Series Convergence Tests**

Evaluate $\int_1^e \ln x \, dx$ using the antiderivative found earlier.

$[x \ln x - x]_1^e = (e \cdot 1 - e) - (0 - 1) = 0 - (-1) = 1$.

18 **[Challenging]** • **Target: Definite integrals using integration by parts** • **Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests**

Evaluate $\int_0^\pi x \sin x \, dx$.

Let $u = x$, $dv = \sin x \, dx$: $\int x \sin x \, dx = -x \cos x + \int \cos x \, dx = -x \cos x + \sin x + C$.
$[-x \cos x + \sin x]_0^\pi = (-\pi(-1) + 0) - (0 + 0) = \pi$.

19 **[Challenging]** • **Target: Integration by parts to derive a reduction formula** • **Cross-Lesson: 6.01 Power Series and Interval of Convergence**

Use integration by parts to show $\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx$ (the reduction formula), and use it once to reconfirm $\int x^2 e^x \, dx$.

Let $u = x^n$, $dv = e^x \, dx$: $du = n x^{n-1} dx$, $v = e^x$. $\int x^n e^x \, dx = x^n e^x - n \int x^{n-1} e^x \, dx$, confirming the reduction formula. With $n = 2$: $\int x^2 e^x \, dx = x^2 e^x - 2 \int x e^x \, dx = x^2 e^x - 2x e^x + 2e^x + C$, matching the earlier result.

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20 [Challenging] • Target: Choosing u and dv wisely • Cross-Lesson: 6.02 Taylor and Maclaurin Series

For each integral, state the best choice of u (using LIATE) without evaluating:

a $\int x^3 \ln x \, dx$ $u = \ln x$ (Logarithmic beats Algebraic)

b $\int x e^{5x} \, dx$ $u = x$ (Algebraic beats Exponential)

c $\int x \tan^{-1} x \, dx$ $u = \tan^{-1} x$ (Inverse trig beats Algebraic)

d $\int x^2 \cos x \, dx$ $u = x^2$ (Algebraic beats Trigonometric)

(a) $u = \ln x$ (Logarithmic beats Algebraic); (b) $u = x$ (Algebraic beats Exponential); (c) $u = \tan^{-1} x$ (Inverse trig beats Algebraic); (d) $u = x^2$ (Algebraic beats Trigonometric)

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: The integration by parts formula

Evaluate $\int_0^{\pi} -2x \cos x \, dx$, using integration by parts.

a. 2

b. 6

c. 4 ✓

d. -4

Let $u = -2x$, $dv = \cos x \, dx$: $du = -2 \, dx$, $v = \sin x$. $\int -2x \cos x \, dx = -2x \sin x - 2 \cos x + C$.
Evaluating: $[-2x \sin x - 2 \cos x]_0^{\pi} = (0 + 2) - (0 - 2) = 4$.

22 [Foundational] • Target: Integration by parts with logarithms and inverse trig

Evaluate $\int_0^{\pi} -6x \cos x \, dx$, using integration by parts.

a. 6

b. 14

c. 12 ✓

d. -12

Let $u = -6x$, $dv = \cos x \, dx$: $du = -6 \, dx$, $v = \sin x$. $\int -6x \cos x \, dx = -6x \sin x - 6 \cos x + C$.
Evaluating: $[-6x \sin x - 6 \cos x]_0^{\pi} = (0 + 6) - (0 - 6) = 12$.

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23 [Foundational] • Target: Repeated integration by parts

Evaluate $\int_0^{\pi} -7x\cos x\,dx$, using integration by parts.

a. 16

b. 7

c. -14

d. 14 ✓

Let $u=-7x$, $dv=\cos x\,dx$: $du=-7\,dx$, $v=\sin x$. $\int -7x\cos x\,dx=-7x\sin x-7\cos x+C$. Evaluating: $[-7x\sin x-7\cos x]_0^{\pi}=(0+7)-(0-7)=14$.

24 [Foundational] • Target: The cyclic case

Evaluate $\int_0^{\pi} 9x\cos x\,dx$, using integration by parts.

a. -16

b. 18

c. -18 ✓

d. -9

Let $u=9x$, $dv=\cos x\,dx$: $du=9\,dx$, $v=\sin x$. $\int 9x\cos x\,dx=9x\sin x+9\cos x+C$. Evaluating: $[9x\sin x+9\cos x]_0^{\pi}=(0-9)-(0+9)=-18$.

25 [Foundational] • Target: Definite integrals using integration by parts

Evaluate $\int_0^{\pi} -x\cos x\,dx$, using integration by parts.

a. -2

b. 4

c. 2 ✓

d. 1

Let $u=-x$, $dv=\cos x\,dx$: $du=-1\,dx$, $v=\sin x$. $\int -x\cos x\,dx=-x\sin x-\cos x+C$. Evaluating: $[-x\sin x-\cos x]_0^{\pi}=(0+1)-(0-1)=2$.

26 [Foundational] • Target: Integration by parts to derive a reduction formula

Evaluate $\int_0^{\pi} -3x\cos x\,dx$, using integration by parts.

a. 3

b. -6

c. 8

d. 6 ✓

Let $u=-3x$, $dv=\cos x\,dx$: $du=-3\,dx$, $v=\sin x$. $\int -3x\cos x\,dx=-3x\sin x-3\cos x+C$. Evaluating: $[-3x\sin x-3\cos x]_0^{\pi}=(0+3)-(0-3)=6$.

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27 [Foundational] • Target: Choosing u and dv wisely

Evaluate $\int_0^{\pi} -4x\cos x\,dx$, using integration by parts.

a. 4

b. -8

c. 10

d. 8 ✓

Let $u = -4x$, $dv = \cos x\,dx$: $du = -4\,dx$, $v = \sin x$. $\int -4x\cos x\,dx = -4x\sin x - 4\cos x + C$.

Evaluating: $[-4x\sin x - 4\cos x]_0^{\pi} = (0 + 4) - (0 - 4) = 8$.

28 [Foundational] • Target: Integration by parts with algebraic-trigonometric products

Given $f(0) = 0$, $f(3) = 2$, $g(0) = 2$, $g(3) = 2$, and $\int_0^3 f'(x)g(x)\,dx = -2$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. 10

b. -6

c. 2

d. 6 ✓

$\int_0^3 f(x)g'(x)\,dx = [f(x)g(x)]_0^3 - \int_0^3 f'(x)g(x)\,dx = [(2)(2) - (0)(2)] - (-2) = 4 - (-2) = 6$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Definite integrals: area under a transcendental product

Given $f(1) = 5$, $f(3) = 6$, $g(1) = 4$, $g(3) = -6$, and $\int_1^3 f'(x)g(x)\,dx = -7$, use integration by parts to find $\int_1^3 f(x)g'(x)\,dx$.

a. -45

b. -49 ✓

c. -63

d. 49

$\int_1^3 f(x)g'(x)\,dx = [f(x)g(x)]_1^3 - \int_1^3 f'(x)g(x)\,dx = [(6)(-6) - (5)(4)] - (-7) = -56 - (-7) = -49$.

30 [Intermediate] • Target: Mixed review: choosing the right technique

Given $f(3) = -1$, $f(6) = 2$, $g(3) = 1$, $g(6) = 3$, and $\int_3^6 f'(x)g(x)\,dx = -8$, use integration by parts to find $\int_3^6 f(x)g'(x)\,dx$.

a. -1

b. 15 ✓

c. -15

d. 19

$\int_3^6 f(x)g'(x)\,dx = [f(x)g(x)]_3^6 - \int_3^6 f'(x)g(x)\,dx = [(2)(3) - (-1)(1)] - (-8) = 7 - (-8) = 15$.

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31 [Intermediate] • Target: Standard Integration by Parts Formula

Given $f(3)=5$, $f(7)=0$, $g(3)=6$, $g(7)=-5$, and $\int_3^7 f'(x)g(x)\,dx=-10$, use integration by parts to find $\int_3^7 f(x)g'(x)\,dx$.

a. -40

b. -20 ✓

c. 20

d. -16

$\int_3^7 f(x)g'(x)\,dx=[f(x)g(x)]_3^7-\int_3^7 f'(x)g(x)\,dx=[(0)(-5)-(5)(6)]-(-10)=-30-(-10)=-20.$

32 [Intermediate] • Target: Tabular Integration by Parts

Given $f(0)=5$, $f(3)=-2$, $g(0)=3$, $g(3)=1$, and $\int_0^3 f'(x)g(x)\,dx=-6$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. -7

b. -11 ✓

c. -23

d. 11

$\int_0^3 f(x)g'(x)\,dx=[f(x)g(x)]_0^3-\int_0^3 f'(x)g(x)\,dx=[(-2)(1)-(5)(3)]-(-6)=-17-(-6)=-11.$

33 [Intermediate] • Target: Cycling Integrals (Solving for the Unknown Integral)

Given $f(3)=-4$, $f(4)=6$, $g(3)=-4$, $g(4)=-6$, and $\int_3^4 f'(x)g(x)\,dx=-2$, use integration by parts to find $\int_3^4 f(x)g'(x)\,dx$.

a. -46

b. -50 ✓

c. -54

d. 50

$\int_3^4 f(x)g'(x)\,dx=[f(x)g(x)]_3^4-\int_3^4 f'(x)g(x)\,dx=[(6)(-6)-(-4)(-4)]-(-2)=-52-(-2)=-50.$

34 [Intermediate] • Target: Integration by Parts with Inverse Trigonometric Functions

Given $f(0)=0$, $f(3)=-6$, $g(0)=-3$, $g(3)=4$, and $\int_0^3 f'(x)g(x)\,dx=-4$, use integration by parts to find $\int_0^3 f(x)g'(x)\,dx$.

a. -28

b. 20

c. -20 ✓

d. -16

$\int_0^3 f(x)g'(x)\,dx=[f(x)g(x)]_0^3-\int_0^3 f'(x)g(x)\,dx=[(-6)(4)-(0)(-3)]-(-4)=-24-(-4)=-20.$

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35 [Intermediate] • Target: The integration by parts formula

Using the cyclic technique, $\int e^{3x}\cos(3x)\,dx = \frac{e^{3x}(3\cos(3x) + C_1\sin(3x))}{18} + C$. Find the coefficient C_1.

a. $\frac{1}{3}$

b. $-\frac{1}{6}$

c. $\frac{1}{6}$ ✓

d. $\frac{7}{6}$

By the standard cyclic-IBP result, $\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$. With $a = 3$, $b = 3$: $C_1 = b = 3$, and the denominator is $3^2 + 3^2 = 18$, so as a fraction of the denominator, $C_1 = \frac{3}{18} = \frac{1}{6}$.

36 [Intermediate] • Target: Integration by parts with logarithms and inverse trig

Using the cyclic technique, $\int e^{5x}\cos(4x)\,dx = \frac{e^{5x}(5\cos(4x) + C_1\sin(4x))}{41} + C$. Find the coefficient C_1.

a. $-\frac{4}{41}$

b. $\frac{4}{41}$ ✓

c. $\frac{5}{41}$

d. $\frac{4}{25}$

By the standard cyclic-IBP result, $\int e^{ax}\cos(bx)\,dx = \frac{e^{ax}(a\cos(bx) + b\sin(bx))}{a^2 + b^2} + C$. With $a = 5$, $b = 4$: $C_1 = b = 4$, and the denominator is $5^2 + 4^2 = 41$, so as a fraction of the denominator, $C_1 = \frac{4}{41} = \frac{4}{41}$.

37 [Intermediate] • Target: Repeated integration by parts

For $\int x^2\tan^{-1}x\,dx$, which is the correct choice of u according to LIATE?

a. $u = \tan^{-1}x$ **(Inverse Trig outranks Algebraic in LIATE)** ✓

b. $u = x^2$ (Algebraic always goes first)

c. It does not matter which factor is chosen for u

d. $u = x^2$ (the opposite LIATE priority)

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u: $u = \tan^{-1}x$ (Inverse Trig outranks Algebraic in LIATE).

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: The cyclic case • Cross-Lesson: 5.02 Ratio, Root, and Alternating Series Tests

For $\int x^2 e^{6x}\,dx$, which is the correct choice of u according to LIATE?

a. $u = x^2$ **(Algebraic outranks Exponential in LIATE)** ✓

b. $u = x^2$ (Algebraic always goes first)

c. It does not matter which factor is chosen for u

d. $u = e^{6x}$ (the opposite LIATE priority)

By LIATE (Logarithmic, Inverse trig, Algebraic, Trigonometric, Exponential), the earlier-listed type should be u: $u = x^2$ (Algebraic outranks Exponential in LIATE).

39 [Challenging] • Target: Definite integrals using integration by parts • Cross-Lesson: 6.01 Power Series and Interval of Convergence

Given $f(1) = 6$, $f(4) = 5$, $g(1) = 4$, $g(4) = 0$, and $\int_1^4 f(x)g'(x)\,dx = -14$, use integration by parts to find $\int_1^4 f'(x)g(x)\,dx$.

a. -6

b. -38

c. 10

d. -10 ✓

Since $\int_1^4 f(x)g'(x)\,dx = [f(x)g(x)]_1^4 - \int_1^4 f'(x)g(x)\,dx$, rearranging:
$\int_1^4 f'(x)g(x)\,dx = [(5)(0) - (6)(4)] - (-14) = -24 - (-14) = -10$.

40 [Challenging] • Target: Integration by parts to derive a reduction formula • Cross-Lesson: 6.02 Taylor and Maclaurin Series

Given $f(0) = 2$, $f(2) = -2$, $g(0) = 0$, $g(2) = -3$, and $\int_0^2 f(x)g'(x)\,dx = -10$, use integration by parts to find $\int_0^2 f'(x)g(x)\,dx$.

a. -4

b. 16 ✓

c. 20

d. -16

Since $\int_0^2 f(x)g'(x)\,dx = [f(x)g(x)]_0^2 - \int_0^2 f'(x)g(x)\,dx$, rearranging:
$\int_0^2 f'(x)g(x)\,dx = [(-2)(-3) - (2)(0)] - (-10) = 6 - (-10) = 16$.

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41 [Challenging] • Target: Standard Integration by Parts Formula • Cross-Lesson: 1.02 Partial Fractions and Improper Integrals

Evaluate $\int xe^{2x}dx$.

a. $\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$ ✓

b. $\frac{1}{2}xe^{2x} + \frac{1}{4}e^{2x} + C$

c. $xe^{2x} - e^{2x} + C$

d. $\frac{1}{2}x^2e^{2x} + C$

Let $u = x, dv = e^{2x}dx \Rightarrow du = dx, v = \frac{1}{2}e^{2x}$. By parts:
$uv - \int vdu = \frac{1}{2}xe^{2x} - \frac{1}{2}\int e^{2x}dx = \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} + C$.

42 [Challenging] • Target: Standard Integration by Parts Formula • Cross-Lesson: 2.01 Euler's Method and Logistic Growth

Evaluate $\int_0^1 x\cos(\pi x)dx$.

a. $-\frac{2}{\pi^2}$ ✓

b. $\frac{2}{\pi^2}$

c. 0

d. $-\frac{1}{\pi}$

Let $u = x, dv = \cos(\pi x)dx \Rightarrow du = dx, v = \frac{1}{\pi}\sin(\pi x)$.
$[\frac{x}{\pi}\sin(\pi x)]_0^1 - \frac{1}{\pi}\int_0^1 \sin(\pi x)dx = 0 + [\frac{1}{\pi^2}\cos(\pi x)]_0^1 = \frac{1}{\pi^2}(-1 - 1) = -\frac{2}{\pi^2}$.

43 [Challenging] • Target: Tabular Integration by Parts • Cross-Lesson: 3.01 Parametric Equations

When is tabular integration applicable?

a. **When one factor is a polynomial that terminates to 0 upon repeated differentiation** ✓

b. Only for trigonometric functions

c. Only for logarithmic integrals

d. When both functions are exponential

Tabular integration is a streamlined form of repeated integration by parts, ideal when one factor is a polynomial that differentiates to zero after finitely many steps.

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44 [Challenging] • Target: Cycling Integrals (Solving for the Unknown Integral) • Cross-Lesson: 3.02 Vector-Valued Functions

Evaluate $\int e^{2x}\sin x\,dx$.

a. $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$ ✓

b. $\frac{e^{2x}}{5}(\sin x - 2\cos x) + C$

c. $\frac{e^{2x}}{4}(2\sin x - \cos x) + C$

d. $e^{2x}\cos x + C$

Integrating by parts twice produces $-\frac{1}{4}I$. Multiplying by $\frac{4}{5}$ yields $\frac{e^{2x}}{5}(2\sin x - \cos x) + C$.

45 [Challenging] • Target: Integration by Parts with Inverse Trigonometric Functions • Cross-Lesson: 4.01 Polar Curves and Derivatives

Evaluate $\int \arctan x\,dx$.

a. $x\arctan x - \frac{1}{2}\ln(1 + x^2) + C$ ✓

b. $x\arctan x + \frac{1}{2}\ln(1 + x^2) + C$

c. $\frac{1}{1 + x^2} + C$

d. $x\arctan x - \ln(1 + x) + C$

Let $u = \arctan x$, $dv = dx \Rightarrow du = \frac{1}{1 + x^2}dx$, $v = x$. Integral is $x\arctan x - \int \frac{x}{1 + x^2}dx = x\arctan x - \frac{1}{2}\ln(1 + x^2) + C$.

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